

1 Collection of Serial Numbers

General

Serial numbers are assigned to be able to differentiate between individual items of material. [Wikipedia](#) provides the following definition for serial numbers:

A **serial number** (also manufacturer's serial number or MSN) is a unique, alphanumeric code assigned for identification of a single unit. Although usually called a number, it may include letters, though ending with digits. Serial numbers are used to identify elements pertaining to series production also providing information about production conditions and thus enabling traceability of the used components, also for electronic products.

Serial numbers can be:

- generated/assigned at the end of the production process
- generated/assigned in earlier stages of the production process and monitored by the process
- integrated as used components in the production process and monitored by the process



[Merging serial numbers](#) represents the assembly of several components listed by serial number into one component part listed by serial number.

MES supports manufacturing businesses in many ways with the collection of serial numbers and enables complete traceability based on the data recorded in MES.

Definition of the supported variants

MES provides different variants to record serial numbers:

Variant	Description of the variant	Description	Documentation
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Variant	Description of the variant	Description	Documentation
1	Entry of serial numbers for OPs that are not subject to management in batches	Recording of serial numbers for operations that are not managed in batches enables the collection in relation to serial numbers if incorporated materials their batches and consumptions are not required to be recorded. The production order specifies the serial numbers to be processed as part of a production order. For each operation the user identifies the serial numbers and classifies them according to the produced quality. The system makes sure that this process can only be performed once for each serial number and operation.	here
2a	Entry of serial numbers for OPs that are subject to management in batches - manual input of the serial number	The user enters the serial number and the system is informed about it. In this case, the serial number is generated in another system. The recorded serial number is assigned a HYDRA batch number.	here
2b	Entry of serial numbers for OPs that are subject to management in batches - automatic assignment of serial numbers	In this case, the system generates the serial number in an unambiguous format applicable to the whole system. Serial numbers are generated according to the rules applying for the generation of batch numbers. In this case, the serial number matches the HYDRA batch number.	here

Variant	Description of the variant	Description	Documentation
2c	Entry of serial numbers for OPs that are subject to management in batches - automatic assignment of serial numbers	In this case, the system generates the serial number in an unambiguous format applicable to the whole system. Serial numbers are generated according to the rules applying for the generation of batch numbers. The serial number is assigned a HYDRA batch number.	Here
2d	Entry of serial numbers for OPs that are subject to management in batches - automatic transfer of serial numbers by using input batches.	In addition, to the conventional generation of output batches, the system particularly supports the generation of merged batches including individually assigned serial numbers. There is a direct connection between individual serial numbers and the generated output batch (merged batch), which logically embraces the serial numbers. Serial numbers are taken over as individual batches provided by logging an input batch/merged batch on. The serial number is the batch number. In addition, individual serial numbers are assigned to a superordinate merged batch.	This document

HYDRA batch number vs. serial number

In HYDRA the [HYDRA batch number](#) is unambiguous throughout the entire system. The serial number is kept throughout the complete process, provided that [serial numbers have not been merged](#). Consequently, it is differentiated between the internal and external HYDRA batch number to represent the collection of serial numbers:

HYDRA batch number

The HYDRA batch number is the ID by which the serial number is managed in HYDRA throughout the complete collection process.

The HYDRA batch number is used on labels to identify the material.

To ensure unambiguity within the system, an internal HYDRA batch number is assigned and managed for every HYDRA batch number when recording serial numbers.

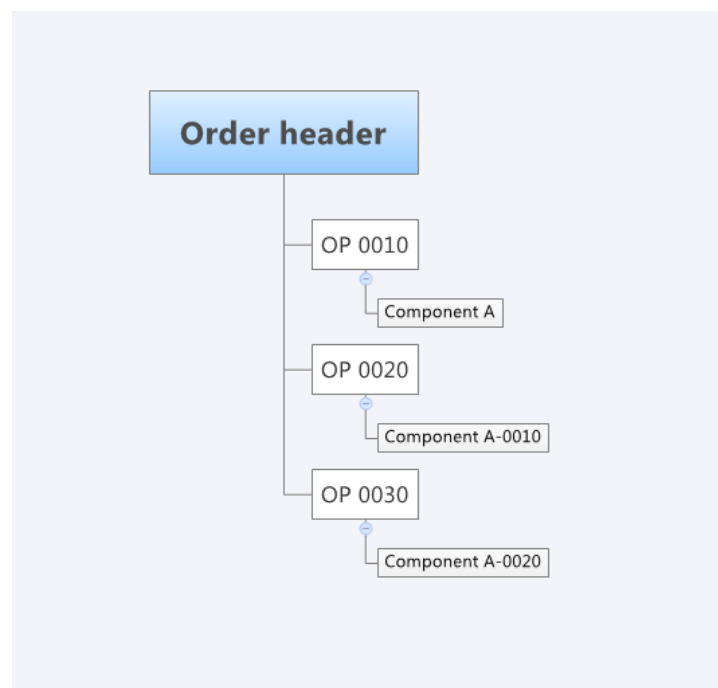
Internal HYDRA batch number

The internal HYDRA batch number represents a system-wide, unambiguous ID identifying batches and serial numbers as an alternative to the HYDRA batch number.

In contrast to the HYDRA batch number; this ID has to be changed from production step to production step to keep unambiguity.

This example explains the procedure:

A production order has the following structure:



The interface provided a merged batch with the serial number "BXC6GF7H". Upon posting this serial number, the HYDRA batch number and the internal HYDRA batch number change as follows:

Process step	Batch status	HYDRA batch number	Internal HYDRA batch number	Comment
Status <u>after</u> transferring the number to MES via the material interface and <u>prior</u> to logging the input batch on.	FREE	BXC6GF7H	BXC6GF7H	Initial status when the HYDRA batch number and the internal HYDRA batch number are identical.
Status after logging the merged batch on to OP 0010	RUNNING	BXC6GF7H	BXC6GF7H	No changes
Status after entering the serial number at OP 0010	Processed	BXC6GF7H	BXC6GF7H	The "old" batch is assigned the "processed" status.
	FREE	BXC6GF7H	PR1111X112	The "new" batch is assigned the "free" status, the same HYDRA batch number (serial number) and a new HYDRA internal batch number
Status after entering the serial number for the OP 0020	Processed	BXC6GF7H	BXC6GF7H	The "old" batch is assigned the "processed" status.
	Processed	BXC6GF7H	PR1111X112	The "old" batch is assigned the "processed" status.

Process step	Batch status	HYDRA batch number	Internal HYDRA batch number	Comment
	Free	BXC6GF7H	PR1111X113	The "new" batch is assigned the "free" status, the same HYDRA batch number (serial number) and a new HYDRA internal batch number
Status after entering the serial number for the OP 0030	Processed	BXC6GF7H	BXC6GF7H	The "old" batch is assigned the "processed" status.
	Processed	BXC6GF7H	PR1111X112	The "old" batch is assigned the "processed" status.
	Processed	BXC6GF7H	PR1111X113	The "old" batch is assigned the "processed" status.
	Free	BXC6GF7H	PR1111X114	The "new" batch is assigned the "free" status, the same HYDRA batch number (serial number) and a new HYDRA internal batch number

As regards a HYDRA batch number (serial number), there might exist 1-n HYDRA batches at different times, whereas only one of the batches may be "free" or "running" at a time.

Please note for the supply via interface

Merged batches and serial numbers can be transferred from other systems (e.g. ERP systems) to MES via the [material staging interface](#). Some particularities have to be taken into account when transferring the merged batch/combination of serial numbers:

- This order has to be observed when transferring serial numbers and their merged batches from Navision to HYDRA:
 1. Transfer of serial numbers
 2. Transfer of the merged batch record
- The below table shows how specific fields are assigned differently when being transferred to HYDRA.

Logical field	Interface field	Merged batch	Serial number
HYDRA internal batch number	HY_LOSNR	Merged batch number specified by ERP	Serial number specified by ERP
HYDRA batch number	DLL	Merged batch number specified by ERP	Serial number specified by ERP
PPS batch	PPS batch	Batch number specified by ERP	Batch number specified by ERP
Merged batch number	MCNR	BLANK	Merged batch number specified by ERP
Merged batch ID	SLOS	Fixed "J" (upper case "j")	BLANK
Merged batch type	SLOSTYP	Fixed "J" (upper case "j")	BLANK

Logical field	Interface field	Merged batch	Serial number
HU level	HULEVEL	1	0
Inventory type	BESTART	Fixed "H"	"S"
Number of individual batches	SLOS	Number of included serial numbers	0

Posting of serial numbers

If a merged batch with assigned serial numbers is finished, the system updates data and generates new documents in the form of log records and goods movements.

Transfer of data from the operation

This data is taken over from the running operation and transferred to the merged batch and the assigned serial numbers:

- Material and material name
- Material type

Transfer of entered data to serial numbers and the merged batch

The machine, producing operation, person performing the posting, material buffer and the transport unit are transferred to the merged batch and the serial numbers included in the merged batch.

Any comment that might be entered is transferred to the merged batch as additional information on the batch.

The status of the merged batch and serial numbers is set subject to the specified quality/class of the output batch:

Selected class	Assigned batch status	Reason accepted
Yield	Free	No
Scrap	Locked	Yes
Rework	Free	Yes

Selected class	Assigned batch status	Reason accepted
Open quantity	Free	Yes

Batch attributes collected in relation to the merged batch are transferred to the merged batch as batch attributes.

Batch attributes collected in relation to the serial number are transferred to serial numbers as batch attributes.

Document links collected in relation to the merged batch are transferred to the merged batch as document links.

Document links collected in relation to the serial number are transferred to serial numbers as document links.

Quantities of the merged batch

The generated merged batch includes the total quantities of the serial numbers assigned to the merged batch.

The number of assigned serial numbers is defined for the merged batch as number of individual batches.

Activities for serial numbers

New batches with a new, internal HYDRA batch number are generated for the serial numbers included in the merged batch (for further details on this procedure, please see section HYDRA batch number vs. serial number) and the preceding batches are assigned the "processed" status.

Generation of ADE log records

An H record is generated for the merged batch. This one includes the quantity(ies) of the merged batch, which in turn result(s) from the quantities of the assigned serial numbers. The batch number of the merged batch is stored at the H record.

Separate log records are not generated for serial numbers.

Generation of goods movements

The generation of goods movements can be controlled individually for consumptions as well as for the output material on the level of serial numbers and merged batches.

Consequently, it is possible to adjust the upload of these goods movements to the conditions of the relevant ERP system.

Details on how to configure goods movements can be found in the [configuration guidelines for serial number recording](#).

Tracing

Tracing information is updated for the merged batch and serial numbers in the system:

- The generated merged batch is connected with all integrated input batches logged on at the time of logging the merged batch on (by logging the OP on or by completing a merged batch beforehand) or that have been logged on since then.
- The serial numbers included in the merged batch are also connected with all integrated input batches logged on at the time of logging the merged batch on (by logging the OP on or by completing a merged batch beforehand) or that have been logged on since then.

[Establishing the connection](#) between serial numbers and input batches can be disabled if required.